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1 Introduction

Epilepsy is a chronic neurological disorder characterized by recurrent, transient episodes of abnormal, synchronous neuronal activity known as seizures. Globally, epilepsy affects approximately 65 million individuals and imposes a substantial social and economic burden, including increased disability and mortality rates (Leonardi and Ustun 2002; Feigin et al. 2021). For the roughly one-third of patients with drug-resistant epilepsy, the unpredictability of seizure occurrence presents a significant challenge in daily life (Schulze-Bonhage and Kühn 2008). The development of reliable, automated seizure prediction algorithms holds the potential to improve patient quality of life by enabling personalized interventions, such as adaptive medication regimens, closed-loop neurostimulation, or timely precautionary actions by patients themselves, thereby increasing patient empowerment and reducing uncertainty.

1.1 Epileptic Cycles and the Proictal State

In seizure prediction, we distinguish between "preictal" states—periods preceding a seizure—and "interictal" states, which are temporally distant from any seizure event (Yang et al. 2024). Numerous studies have demonstrated that machine learning models can distinguish these states with performance significantly better than chance in both human and canine intracranial EEG (iEEG) recordings (Brinkmann et al. 2016; Kuhlmann, Karoly, et al. 2018). Notably, Cook et al. (2013) provided evidence that, in some cases, seizures can be predicted hours in advance.

A growing body of research has revealed that seizures and related interictal epileptiform activity (IEA) are not randomly distributed in time, but instead exhibit rhythmicity across multiple temporal scales, including circadian (daily), multidien (multi-day), and circannual (yearly) cycles (Karoly et al. 2021). Seizure occurrence is often phase-locked to specific phases of these cycles, suggesting that the brain's susceptibility to seizures fluctuates over time.

Despite these insights, the mechanisms underlying ictogenesis—the process by which a normal brain transitions into a seizure—remain incompletely understood (Kuhlmann, Lehnertz, et al. 2018). A central question is whether seizures arise spontaneously, with little or no warning (i.e., within seconds to milliseconds), or whether the brain undergoes a gradual transition into a state of heightened seizure susceptibility over minutes, hours, or even longer. The latter scenario is conceptualized as the "proictal" state, in which the brain is primed for seizure generation and only a minor perturbation is required to trigger a seizure. In this framework, certain phases of the aforementioned cycles correspond to periods of increased proictal risk.

In the context of seizure prediction, two major challenges persist: (1) understanding if or why some seizures are not preceded by detectable preictal changes and thus remain unpredictable (sensitivity), and (2) elucidating the origins of false positive predictions (specificity). One hypothesis is that not all seizures are preceded by reliable preictal signatures, reflecting truly

spontaneous ictogenesis. Alternatively, limitations in model architecture or feature selection may prevent the detection of existing preictal patterns. A further possibility is that false positive predictions may, in fact, correspond to periods when the brain is in a proictal state—i.e., a seizure was likely, but the critical perturbation did not occur, resulting in a subclinical event (Müller et al. 2022).

1.2 Research Approach

In this work, we systematically investigate these possibilities. Two distinct model types are trained: a convolutional neural network (CNN) that processes raw iEEG signals, and an ensemble of multilayer perceptrons that operate on hand-crafted features extracted from the signals. The temporal co-occurrence of false positive predictions across these models is interpreted as evidence for the existence of a proictal state. Additionally, we analyze whether false positive predictions are phase-locked to the same cycles as seizures. If false positives cluster in the same cycle phases as true seizures, this further supports the hypothesis that these periods represent genuine proictal states, even in the absence of overt seizures.

2 Methods

2.1 Dataset Description

Data from two different datasets called "Ultra2" and "Competition" are used. Patients are referred to with the initial letter of their dataset, followed by their ID number, e.g., U17 or C03.

Both datasets consist of seizure annotations and electroencephalography (EEG) data, stored as European Data Format (EDF) files. The seizure's starts, and in some cases the type and end, were contained in the annotations.

2.1.1 Ethics Votum

2.1.2 Device Description

2.1.3 Origin of Seizure Annotations

2.1.4 Inclusion Criteria

For a patient to be included in this study, generally speaking, they needed to have enough EEG data and seizures. These were necessary, so the models have enough examples to learn features which can be used to forecast seizures and for the evaluation to carry statistical weight. Only patients who had completed the U002 study at the point of writing and thus whose entire EEG data was available were considered.

Patients needed to have at least 10 valid seizures. In this context, a seizure is considered valid if no other seizure occurs within its preictal interval (section 2.4.3), i.e., a seizure must occur at least ~65 min after the previous seizure.

Regarding the recording duration, there are two separate concepts. First, there is the total monitoring timespan, which is the duration from the start of the first recording to the end of the last. However, data is not recorded all the time, since the device is regularly taken off for a while. So secondly, the total duration of all available recordings is taking into account. Based on these, we want to quantify how much of the monitoring timespan data is available. We define this so-called **recording coverage** as:

$$\text{recording coverage} = \frac{\text{total recorded duration}}{\text{total monitoring timespan}}$$

Patients with a recording coverage less than 60 % were excluded. No patients needed to be excluded on the basis of the total recorded duration, as no patients with little EEG data had enough seizures.

Table 2.1 shows which patients were included and why.

Patient		Seizures			Timespan			
ID	Incl.	Total	Valid	Met	Monitoring	Recorded	Coverage	Met
C01	✓	92	58	✓	494	388	78 %	✓
C02	✓	54	53	✓	537	487	91 %	✓
C03	✓	63	63	✓	415	394	95 %	✓
U01	✓	259	195	✓	401	381	95 %	✓
U02	×	2	2	×	293	102	35 %	×
U03	✓	59	52	✓	394	295	75 %	✓
U04	✓	64	62	✓	197	180	91 %	✓
U05	✓	76	68	✓	358	299	84 %	✓
U06	×	0	0	×	93	84	90 %	✓
U07	✓	13	13	✓	132	124	94 %	✓
U08	×	9	7	×	179	137	76 %	✓
U09	×	0	0	×	301	291	97 %	✓
U10	×	0	0	×	201	165	82 %	✓
U11	×	0	0	×	216	155	72 %	✓
U12	✓	37	30	✓	309	199	64 %	✓
U13	×	4	4	×	202	115	57 %	×
U14	×	5	3	×	420	398	95 %	✓
U15	✓	30	23	✓	352	308	88 %	✓
U16	✓	16	16	✓	346	222	64 %	✓
U17	✓	21	21	✓	264	248	94 %	✓
U18	×	4	4	×	210	188	89 %	✓
U22	×	3	3	×	3	3	89 %	✓

Table 2.1: Patient Inclusion: shows which inclusion criteria were met for which patients. Next to the valid seizures ($\geq$ 10) and recording coverage ($\geq$ 60 %) conditions is denoted whether they were met.

2.2 Dataset Cleaning

The initial datasets had many formatting inconsistencies and erroneous data. This section will outline which issues were found and how they were dealt with.

2.2.1 Seizure Annotations

The annotations of the Ultra2 dataset were initially stored as raw text files, generally with one file per combination of patient and visit.

These files suffered from inconsistent formatting and naming, typos, and files being placed in the wrong patient's directory. To systematically work with the files, these issues were rectified, and they were converted to a comma-separated values (CSV) format for clearer structure and combined into a single file per patient.

Patient U03 had seizure annotations during a period which was no longer available, as the data was lost during the study. These annotations were removed.

2.2.2 EEG Recordings

For some of the EEG files, the following systematic issues were identified:

1. The file was corrupted.
2. The file was labeled as "unknown implant" and contained only artifacts.
3. The file reported a start date in the early 2000s, i.e., before the start of the study.
4. Duplicate copies of a patient's files were found in another patient's directory.
5. Duplicate copies were present within a patient's directory.
6. Multiple files shared the same start timestamp within a patient's directory.

Files with issues 1–3 were removed. For files with issues 4 and 5, only a single copy in the correct directory was retained. For files with issue 6, the longest file was retained; if multiple files had the same length, one was selected at random.

Additionally, it was verified that all EDF files have the same sampling frequency, the same channel names in the same order, and all annotated seizures were within recordings.

2.3 Overview of the Processing Pipeline

2.4 Preprocessing

2.4.1 Timestamp Localization

EDF files store their start as a naive timestamp, i.e., without information about their time zone. Daylight saving time (DST) caused the timestamps to have different UTC offsets within a patient.

To rectify this, the timestamps for a patient were all converted to that patient's standard time — referred to as their "main" time zone — based on their locale.

The challenge when doing this is that the time shift backwards in autumn creates ambiguous times during the hour that occurs twice. These timestamps' UTC offset cannot be inferred without an ordered sequence of timestamps in which the time jumps back by one hour at some point.

To achieve this for the competition patient's files, the files were sorted by the sequence of recording, and then the file's start and ends were put in order in the format *file1 start, file1 end, file2 start, file2 end, etc*. This allowed the detection of where the end of the previous file was later than the start of the current file, which indicated a time shift backwards, i.e., the shift from DST to standard time. This allowed for finding the border of which files were in daylight saving time and their appropriate conversion to standard time.

For Ultra2 patients, the EDF files contained additional information about their time zone, which was used for conversion.

2.4.2 EEG Artifact Removal

There are two types of artifacts that occur frequently in the EEG data. Their amplitude is very extreme compared to the rest of the signal; thus, they would disrupt multiple parts of the further processing, such as model training and cycle extraction, making it important to remove them.

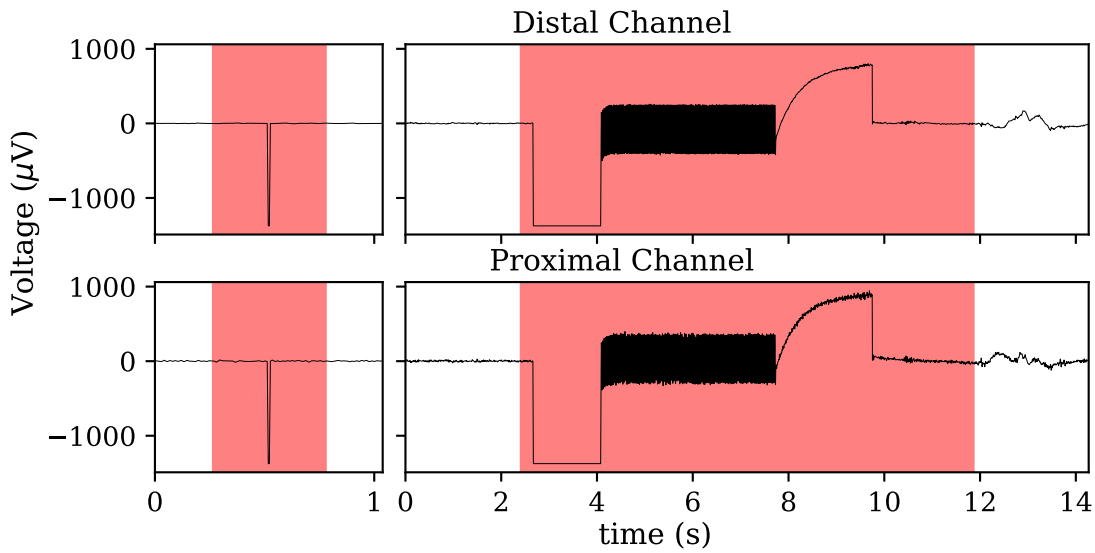

Figure 2.1: Typical EEG Artifacts. Removed intervals are shown in red. Left: A short artifact where the signal is clipped for a single sample. Right: A longer artifact where the clipping is followed by rapid oscillation and a DC drift.

As shown in fig. 2.1, artifacts usually occur in both channels. The signal hits its minimum possible amplitude of $-1374.21\,\mu\text{V}$.

In the first type of artifact, it just clips for one or more samples, as shown in the left panel of

fig. 2.1. In the second type of artifact, the clipping is followed by rapid oscillation, then a DC drift before stabilization back to the clean signal, as shown in the right panel of fig. 2.1.

To filter out these artifacts, first, all signals are scanned for clipped intervals where there are samples that are within 0.1 µV of the most extreme possible values on either channel. After the end of each clipped interval, the variance is calculated in a two-second window overlapping with a 0.5 sec step. As long as consecutive window's variances are above a threshold, they keep getting removed. The high-variance check captures the oscillations as well as the DC drift. Multiple variance thresholds were tested, and a threshold of 20 000 µV was determined to reliably catch these artifacts without removing valid data. Care was taken not to remove too much high-variance signal, as this often occurs during seizures, which are of particular importance to the analysis. The detected invalid intervals were then padded with 0.25 sec on both sides to ensure that the adjacent data, which might also have been contaminated, are removed. Finally, intervals were merged if they were less than a segment's length apart, since that is the shortest window used here.

This resulted in a set of invalid recording intervals, which were excluded from the further analysis, as well as a set of valid intervals, which were included.

2.4.3 Windowing

For downstream parts of the pipeline, we create two types of window in the recordings: short *"segments"* and longer *"clips"*. In this paper, these terms carry the specific meanings that are defined in this section.

Segment Duration

Segments are the shortest windows that are used. Based on Yang et al. (2024), they are approximately 15 sec long. However, the EEG data here has an unusual sampling frequency of 207.031 054 658 198 7 Hz, which for a 15 sec segment would result in a fractional number of samples. To avoid this, the exact segment duration was changed to the closest duration that resulted in an integer number of samples which resulted in a duration of 14.997 75 sec with exactly 3105 samples.

The timespan from the start of the first valid interval to the end of the last is completely filled with contiguous, non-overlapping segments, resulting in an aligned series of segments without gaps. Segments that are fully contained in valid intervals are labeled as "existing" and segments that are only partially or not contained are considered non-existing and are not further considered.

Segments receive a "type" which indicates their position relative to close seizures. This type is determined based on the end of a segment, as predictions are issued at the end of clips, which are based on the end of segments.

Interval Types

For seizure prediction to be useful, it is necessary to have a certain minimum intervention interval between the warning time, when the prediction is issued, and the seizure's start. This would enable the administration of an inhibitory drug or the patient to take cautionary action. On the

other hand, for a warning to be useful, the seizure must occur within a certain time frame after the intervention time has passed. The window in which the seizure is expected to occur after the intervention time is called the seizure prediction horizon (SPH).

The equivalent to the SPH when looking backwards in time from a seizure, rather than forwards in time from a prediction, is called the preictal interval. It is the window before a seizure where a warning should be issued (fig. 2.2).

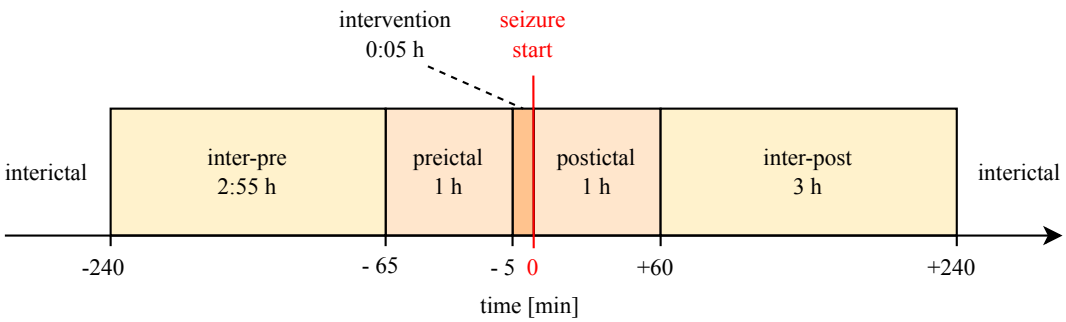

Figure 2.2: The intervals around a seizure with their approximate durations and relative time points. *Not to scale.*

Additionally, it is important to take into account that around the seizure, even beyond the preictal interval, the signal properties might be changed. The period before the preictal interval is labeled as "inter-pre". Similar to the preictal interval, an interval after a seizure is called "postictal" and the period after that is called "inter-post". Periods that do not fall within any of these intervals are labeled as "interictal".

The exact durations of the intervals are calculated as a multiple of the segment duration, again due to the unusual sampling rate (table 2.2). This facilitates analysis by letting each interval have a whole number of segments.

Interval	**Approximate duration [h]**	**Segments contained**	**Exact duration [h]**
segment	0:00:15	1	0:00:14.99775
clip	0:10:00	40	0:09:59.91000
inter-pre	2:55:00	700	2:54:58.42500
preictal / SPH	1:00:00	240	0:59:59.46000
intervention	0:05:00	20	0:04:59.95500
postictal	1:00:00	240	0:59:59.46000
inter-post	3:00:00	720	2:59:58.38000

Table 2.2: The approximate and exact durations of the defined intervals, as multiples of segments.

Clips

As in Yang et al. (2024), non-overlapping clips that are approximately 10 min long are created. As with the other intervals, their exact duration is adjusted to a multiple of a segment (table 2.2). The clips are aligned to the segments and usually contain 40 thereof, regardless of whether they are existing segments. They inherit their type from the segments they contain, which means they can have mixed types. Since clips are used for later evaluations, they should not contain a mix of preictal and non-preictal segments, as they could not be clearly labeled then. For this reason, they are also aligned to the ends of preictal intervals, such that one clip always ends at the end of a preictal interval, and further clips are created looking backwards in time from there.[1] This means each preictal interval will be fully filled with six clips. Hence, the algorithm that creates the clips takes the end of each preictal interval, and the end of the last valid recording interval as seeds, and creates clips looking backwards in time from there. However, the duration between seizures is usually not divisible by 10 minutes without remainder, and therefore the duration between the ends of preictal intervals is neither. This means that the number of segments between preictal intervals is not usually divisible by 40, and therefore, when looking backwards, the last clips before the next preictal interval will usually be shorter than 40 segments.

This results in two possible issues for a clip:

1. It contains too few segments, whether they exist or not.
2. It contains too few existing segments, i.e., it is not sufficiently located in a valid recording interval.

Clips that do not meet these criteria are considered invalid and excluded from parts of the later procedure. Regarding the first issue, a clip must contain the full 40 segments to be valid. Regarding the second issue, a clip must contain at least 35 existing segments to be valid. This lets clips be that contain short artifacts or whose edge falls outside a recording, but excludes clips where a significant portion is missing.

2.4.4 Dataset Partitioning

We want to train the models on a subset of the data and evaluate their performance on unseen data. To achieve this, the data is split into a train and test set. Since predictive models must use past data to predict future events, the time series is split at a certain time point, rather than allocating individual clips to the sets. An additional reason for this is that we evaluate cycles of up to 50 days, which requires long stretches of continuous data.

Generally speaking, for partitioning, it is important that the train set contains enough samples of each class to learn the patterns and the test set must contain enough samples to accurately evaluate the performance. Here, to evaluate cycles, the test set must also be long enough to include at least the longest cycle.

The classes here are preictal and non-preictal. Since seizures only create ~1 h of preictal data but only tend to occur every few days or less, there is a large class imbalance in the data with

[1] Since this is also done for the test set, it is one of the reasons why the performance evaluation cannot be considered purely prospective as with unseen data.

scarcity of preictal and an abundance of non-preictal data – only 0.76 % of all patient's clips are preictal. Hence, it is important to consider the number of seizures per split, in addition to the recording duration. We aimed to allocate ~60 % of seizures for training and ~40 % for testing.

The train-test boundary was determined per patient as a timestamp on the clip grid. First, seizures were ordered chronologically and an initial split was targeted at 60 % training seizures and 40 % test seizures. If N denotes the number of valid seizures, the number initially allocated to training was $\lfloor 0.6N \rfloor$. The approximate split timestamp was set to the midpoint between the last training seizure and the first test seizure. This timestamp was then aligned to the clip representation by snapping it to a clip start.

Next, the split was constrained using valid clips to ensure adequate EEG data in both partitions: the train partition had to contain between 40 % and 70 % of valid clips. If the snapped split violated these bounds, the split position was shifted in clip-index space to the nearest admissible bound. Finally, the boundary was required to start at an interictal clip; if necessary, the closest interictal clip was selected (searching forward or backward depending on which bound adjustment was needed). The start time of this clip was used as the final boundary.

Patient U03 had a large gap in the middle of his monitoring timespan where data was lost during the study. Following the partitioning procedure, the boundary would have fallen slightly before this gap, creating a very short portion of data, followed by a large gap. This hindered the cycle evaluation; therefore, the boundary was manually set to fall into the gap for this patient.

2.4.5 Feature Extraction

The features extracted from the signals per segment as shown in table 2.3. Some of these were used as inputs to the feature-based multilayer perceptrons (FB-MLPs), and all were used for the evaluation of cycles.

Feature	#Values per channel	#Values	Used for ensemble
Correlation coefficient	-	1	✓
ACFW	1	2	
Variance	1	2	
Bandpowers	5	10	✓
Total		15	

Table 2.3: Features and their properties

We computed the signal variance and autocorrelation function width (ACFW) based on Maturana et al. (2020), and Pearson's correlation coefficient between the two channels, and the spectral band powers of each channel's signal based on Müller et al. (2022).

The ACFW was calculated for each channel. The autocorrelation function (ACF) represents the correlation of a signal s with itself shifted by a lag value λ, which corresponds to a number of samples. $\bar{s}$ is the signal mean, T is the number of total samples, Var_s is the variance of s.

$$\mathrm{ACF}_\lambda = \frac{1}{T} \frac{\sum_{t=1}^{T-\lambda} (s_t - \bar{s})\,(s_{t+\lambda} - \bar{s})}{\mathrm{Var}_s} \tag{2.1}$$

The ACF is bounded between -1 and 1. The ACFW is defined as the lag λ where the ACF reaches its half-maximum value of 0.5. It can be interpreted as a measure of temporal persistence: it quantifies how quickly a signal decorrelates over time.

Lastly, the spectral band powers were calculated for the delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (12–35 Hz), and gamma (35–100 Hz) bands for each channel. Band powers were computed for each segment and channel from the power spectral density (PSD) estimated with Welch's method.

Specifically, the signal was divided into Hann-windowed subwindows of length $(2/f_{\min})$ (where $f_{\min} = 0.5$ Hz is the lower bound of the lowest spectral band), with 50 % overlap, detrending by subtraction of the mean, and density scaling (resulting in the unit $V^2/r\text{Hz}$). This yielded a PSD over frequency for every segment-channel pair.

For each predefined band, the absolute band power was then obtained by numerically integrating the PSD between the band limits using Simpson's rule, with the frequency-bin spacing as the integration step. It was important to use the absolute, rather than relative band power, so that it stays comparable between segments. The result is one scalar band-power value per band and per channel (10 values total for two channels and five bands).

2.5 Model Architectures

In this project, a CNN and an ensemble of FB-MLPs were trained per patient to predict seizures. This section will discuss their exact specifications.

2.5.1 Convolutional Neural Network

In this project, the architecture of the CNN was adapted from Müller et al. (2022) which used the nv1x16 topology originally proposed in Eberlein et al. (2018). It analyzes a segment and uses the raw EEG signals per EEG channel as inputs. Until the final stages, it treats EEG channels separately, computing features per channel. Only at the end are the per-channel features combined to output a final score.

In the following, EEG channels refer to electrode-derived input signals, whereas CNN channels refer to convolutional feature maps (tensor depth).

Figure 2.3 shows the exact architecture. The inputs are passed through a combination of convolutional and pooling layers. There are two distinct types of convolutional blocks: conv1 and conv2. Conv1 uses a convolutional layer, followed by a pooling layer with a relatively large kernel size, reducing the height (#samples) rapidly. All convolution layers use L1 and L2 regularization with a penalty of 10^{-9}. Four conv1 blocks are followed by three conv2 blocks. These further reduce the height by using valid padding for two convolutional layers and pooling again. Ultimately, the sample dimension (height) is reduced to 1 with 32 features per EEG channel. Only then are the per-channel features combined (flattened). These pass through two final dense layers to create the output logit, which is then bounded between 0 and 1 with the sigmoid function, to create the final output.

Since the height gets reduced further and further in deeper layers, the extracted features receive information from more samples, i.e., the receptive field of deeper neurons grows on the time

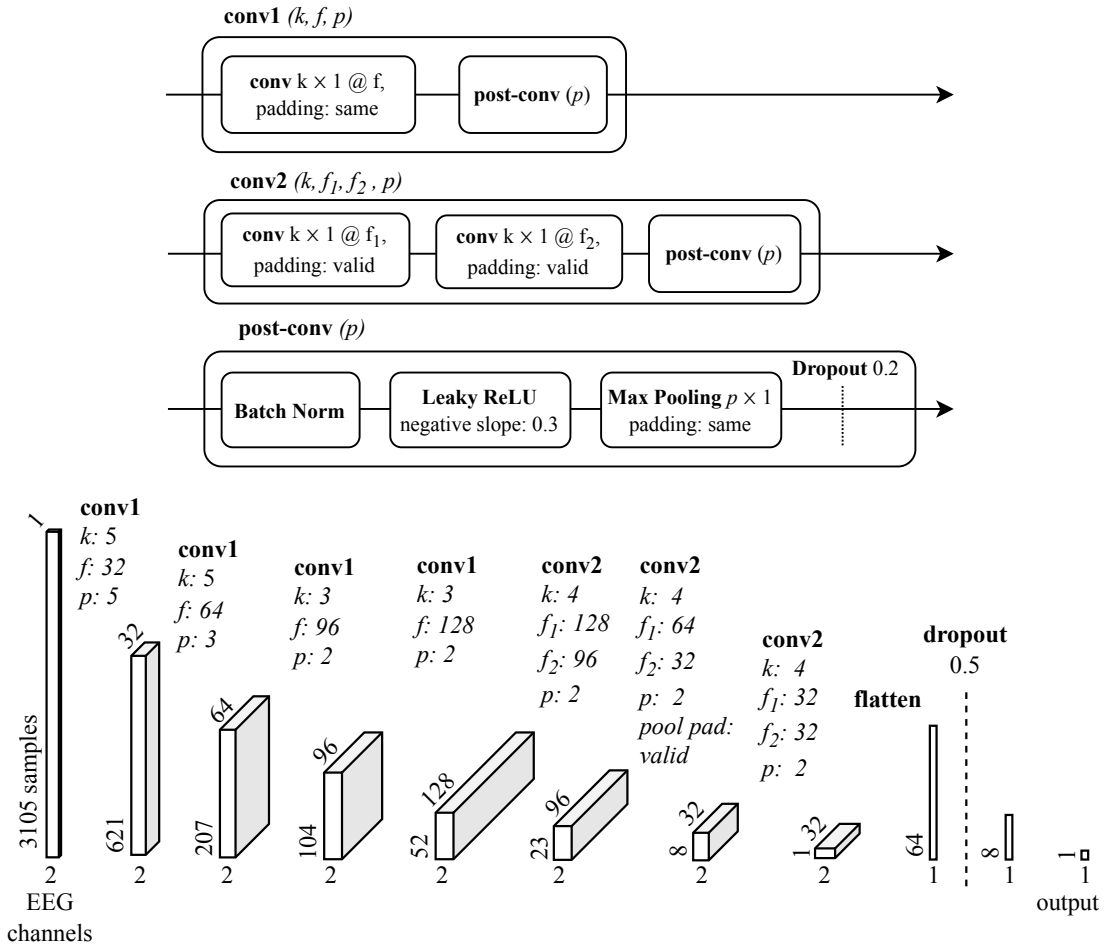

Figure 2.3: CNN Architecture.
Top: Types of special blocks where k: kernel size f: #filters, p: pool size.
Bottom: A segment's EEG data is processed from left to right. The volumes represent the input tensor and the outputs of layers. In between the volumes, the network's layer's properties are shown.

scale. Since the per-channel features are only combined after the final conv block, information from the EEG channels is evaluated separately until the end.

To accommodate two differences in the data structure, compared to Eberlein et al. (2018), two adjustments were made to the architecture. Here, 2 EEG channels are used instead of 16. This means there are 8 times fewer nodes in the first flattened tensor. For this reason, the number of nodes in the first dense layer was reduced by a factor of 8: from 64 to 8. The flat tensors thus have a height of $64 \rightarrow 8 \rightarrow 1$, instead of $512 \rightarrow 64 \rightarrow 1$.

Secondly, because the sampling rate of ~207 Hz here differs slightly from the sampling rate of 200 Hz, the number of input samples from a segment differs too (3105 vs. 3000). Thus, to arrive at a height of 1 at the final conv layer, it is necessary to reduce this dimension slightly more. For this reason, the pool padding of the second conv2 layer was changed from same to valid. This minor change is enough to fully accommodate the sampling frequency difference.

2.5.2 Ensemble of Feature-Based Multilayer Perceptrons

The second model is an ensemble of FB-MLPs that was also adapted from Müller et al. (2022). Instead of raw signal, it uses hand-picked features: specifically the band power spectrum and correlation coefficient from section 2.4.5. This allows it taking features of individual channels and their correlation into consideration.

An individual FB-MLP uses the z-score normalized features as its input layer. It has three dense, hidden layers comprising 16, 8, and 4 nodes, which use recfitied linear unit (ReLU) as an activation function. Batch normalization follows the input and hidden layers. The output layer uses the sigmoid activation function.

The ensemble consists of 50 FB-MLPs that are initialized with different random weights and share their input layer. The outputs of the individual models are then averaged to get the final output. Müller et al. (2022) use an ensemble size of 100, rather than 50. However, they also use many more features as inputs, which enables greater variability between models. Since the variability cannot be as high here, a smaller ensemble size was chosen to reduce resources needed for training.

2.5.3 Model Training

To train the models, only preictal and interictal segments were used. Inter-pre and intervention segments were not used as are shortly before a seizure and thus might often contain similar features to preictal segments. If this was the case, they would make the model learn the wrong patterns since they are still labeled as 0 (not preictal). Postictal and inter-post segments were not used because they might be in the preictal window of non-valid seizures and thus contain preictal features. Additionally, post-seizure, the brain state can be altered significantly, causing changes in the signals, which could confuse the models (Müller et al. 2022).

As mentioned in section 2.4.4, there is a large class imbalance with scarcity of preictal segments. To not waste resources training on more interictal segments than necessary, we subsampled the number of interictal segments to max. 20 times the number of preictal ones. The class weights were adjusted to the final ratio of preictal vs. interictal segments, so that the models

were not biased towards focusing mostly on learning to predict interictal segments. All models received the same subsampled segments for training, so that results are more comparable.

As is Eberlein et al. (2018), the CNNs were trained for 50 epochs with ADAM optimization, a learning rate of 0.001, and binary cross-entropy as a loss function.

To train the FB-MLPs, before subsampling, a z-score normalizer was fit to the features of the entire train set. This same scaling was then always applied when this model was later used, also for the test set. They were then trained for 200 epochs using stochastic gradient descent for optimization, a learning rate of 0.0001, and binary cross-entropy as a loss function. Müller et al. (2022) used 500 epochs for training. However, here the performance could not be seen to improve after 200 epochs, which is likely also due to the smaller input dimension, requiring less fitting.

2.6 Model Evaluation

After training, it was important to evaluate the performance of the models on the valid clips, as conclusions about the propensity of seizures could only be drawn from the false positives (FPs), if the models were decent at predicting seizures. FPs from a random or worse-than-chance classifier would not carry the intended meaning.

For the ensemble, the input features were scaled again using the z-score normalizer, which was previously fit on the train data. Then, the models were used to issue scores (0–1) for all existing segments, which are the model's prediction of how likely that segment is to be preictal. The segment scores were then averaged per clip to get the clip scores. These, rather than segment scores, are used for the further evaluations.

We used two types of metrics to evaluate the performance: clip-based metrics and event-based metrics (EBMs).

Here, clip-based metrics refer to the standard measures used to evaluate models in machine learning on samples (clips), which use the confusion matrix of true positives (TPs), FPs, false negatives (FNs), and true negatives (TNs) to compute metrics, such as precision, sensitivity, accuracy, and the area under the curve of the receiver operating characteristic (ROC AUC). The Hanley McNeil test (Hanley and McNeil 1983) was performed to test for statistically significant improvement of the ROC AUC compared to a random classifier.

On the other hand, EBMs are specific to time-series data and seizure prediction and quantify more clinically relevant properties. Here, the (relative) time in false warning (TIFW) (Mormann et al. 2007; Eberlein 2025) and the (relative) number of seizures predicted were used.

2.6.1 Time in False Warning

Relative TIFW is the event-based (EB) equivalent to the false positive rate (FPR), i.e., the inverse of specificity. That is to say it is minimized when non-preictal (false) samples are correctly classified as such. However, when a prediction is issued from a ~10 min clip, it is not issued for the timespan of that clip, but for that clip's SPH, i.e., the timespan ~5–65 min after the clip's end. This means that summing the duration of FP clips is not sufficient to calculate the TIFW. The clip's SPHs and overlaps thereof must be taken into account. When a system unnecessarily

alarms the patient, it causes them distress and inconvenience, which is why it is essential to minimize false alarms.

To calculate TIFW, only valid, non-preictal clips are selected. For all such clips that issue a warning (FPs), the SPHs are selected. Overlapping SPHs are then merged, so that shared time is not counted multiple times. The total duration of these non-overlapping false-warning intervals is the absolute TIFW. The relative TIFW is obtained by normalizing absolute TIFW by the maximum possible warning time, which is the sum of the non-overlapping SPH intervals from all non-preictal clips. This yields a metric bounded between 0 and 1 which enables comparable interpretation across patients and is consistent with the guidelines in Mormann et al. (2007).

2.6.2 Event-Based Sensitivity

It is also essential to quantify the relative number of seizures successfully predicted, which is also referred to as EB sensitivity. For preictal clips, a positive prediction is considered a true warning, and a seizure is counted as predicted if it falls within the SPH of at least one positively predicted preictal clip. This yields the number of predicted seizures, and EB sensitivity is then computed as the ratio of predicted to total seizures. This metric therefore reflects clinically relevant event detection performance at seizure level rather than clip level.

2.6.3 Threshold Optimization

Whether a clip issues a warning depends on whether its score is greater than or equal to a classification threshold. To optimize it, metrics are calculated for all possible thresholds, i.e., all unique model scores. To take both the EB specificity (1 - rel. TIFW) and the EB sensitivity into account, the harmonic mean between them is calculated:

$$H(a, b) = \frac{2ab}{a + b} \tag{2.2}$$

$$S_{\text{EB}} = H\big(\text{Sensitivity}_{\text{EB}},\ 1 - \text{rel. TIFW}\big) \tag{2.3}$$

The threshold that maximizes this EB score is set as the best threshold and used to determine binary clip predictions in further steps.

The metrics are calculated for the train and test set independently. The results for the test set yield a more accurate evaluation of the performance, since it is unseen data. However, the optimal threshold is determined for the test set individually, rather than using the train set's threshold. This again cannot be considered purely prospective performance, but improves the performance, which is an advantage for the cycle extraction, and the priority here.

2.7 Model Comparison

If different models both classify a portion of data as preictal, even though no seizure occurs afterward, this is interpreted as evidence for a proictal state, since preictal features were apparently present in the data. To quantify the tendency of false predictions to co-occur across models, we calculated the correlation of prediction errors (COPE) (Müller et al. 2022) between the ensemble and CNN scores for each patient. Its definition follows.

Let $\vec{p}_i$ and $\vec{p}_j$ be the model's outputs for models i and j, respectively, and $\vec{L}$ the correct label vector. The prediction error vector of model i is defined as $\vec{e}_i = |\vec{p}_i - \vec{L}|$ (analogously for model j). The weight w_k for each sample k, the weighted mean $\langle z \rangle_w$, and weighted standard deviation $\sigma_w(z)$ are defined as:

$$\begin{aligned} w_k &= \max(e_{i,k}, e_{j,k}), \\ \langle z \rangle_w &= \frac{\sum_k w_k z_k}{\sum_k w_k}, \\ \sigma_w(z) &= \sqrt{\langle z^2 \rangle_w - \langle z \rangle_w^2}. \end{aligned} \tag{2.4}$$

The COPE is then

$$c_w = \frac{\left\langle (\vec{e}_i - \langle \vec{e}_i \rangle_w)\,(\vec{e}_j - \langle \vec{e}_j \rangle_w) \right\rangle_w}{\sigma_w(\vec{e}_i)\,\sigma_w(\vec{e}_j)}. \tag{2.5}$$

This measure emphasizes coherence in false predictions by assigning higher weight to samples where at least one model has a large prediction error. Consequently, samples with low errors contribute little, whereas samples with high errors dominate the estimate.

2.8 Cycle Extraction

2.8.1 Gap Filling

Cycle extraction requires gap-free data, so gaps are filled before processing. This is done on the segment level because more data are available there than at the clip level. Otherwise, segment data from invalid clips would be lost.

The gap-filling procedure preserves the original segment timeline while imputing only short missing intervals ($< 14\,\mathrm{d}$) in the extracted feature series. Long gaps ($\geq 14\,\mathrm{d}$) are treated as structural recording interruptions and are therefore not imputed. Instead, they are used to split the data into independent continuous chunks, ensuring that donor values are never drawn across major recording breaks.

Within each fillable gap, imputation is performed independently per feature column by local donor sampling. For a gap of length L, donor candidates are taken from neighboring segments before and after the gap within a window of size $W = \max(L, N_{\min})$, where $N_{\min} = 100$ is the minimum donor span. Values are sampled with replacement from valid donor rows and assigned to the missing positions. Additionally, each sampled value is perturbed by a multiplicative factor in the interval $[1 - \delta,\ 1 + \delta]$, where $\delta = 0.5$ is the maximum distortion percentage. This design preserves local feature statistics, avoids unrealistic interpolation over long outages, and introduces slight stochastic variation to reduce deterministic copy artifacts in the imputed sequences.

2.8.2 ...

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